Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=5;i++)
    (i-1)[arr] = (i%5);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 3;
}
int main(){
  int arr[6] = {2,2,2,2,2};
  func1(arr+1);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

2, 1, 2, 3, 4, 5

arr = 2

a[1] = & arr[0]

a[0] = & arr[3]

Output
212340
212340

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 4;
  a = &temp;
}
```

Explain your answer

No, because it is not affecting (arr) array. It's kind of pointer of pointers. ~~The~~ It is only playing with pointers ~~not with~~ their values

Q. 2: Write the output of the following sequence of instructions **long arr2[90]; int arr1[80], a1=(&arr2[60]), a2=(&arr2[10]), b1=(&arr1[60]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

~~60~~ 50 50

This is because arrays are stored in continuous memory locations. So, if we subtract their address ~~at the~~ ~~address~~ ~~answer~~ answer will be 50 50 because adjacent memory locations differ by 1.

(0, 1, 2, 3, 4)

2, 1, 2, 3, 4, 5 [arr] = 1

2, 4[arr] (1, 2, 3, 4, 0)

2, temp = &arr[4]